Arjun Arvind Gahane (25070521135)

Exp 14.1

Algorithm:

Step 1: Start.

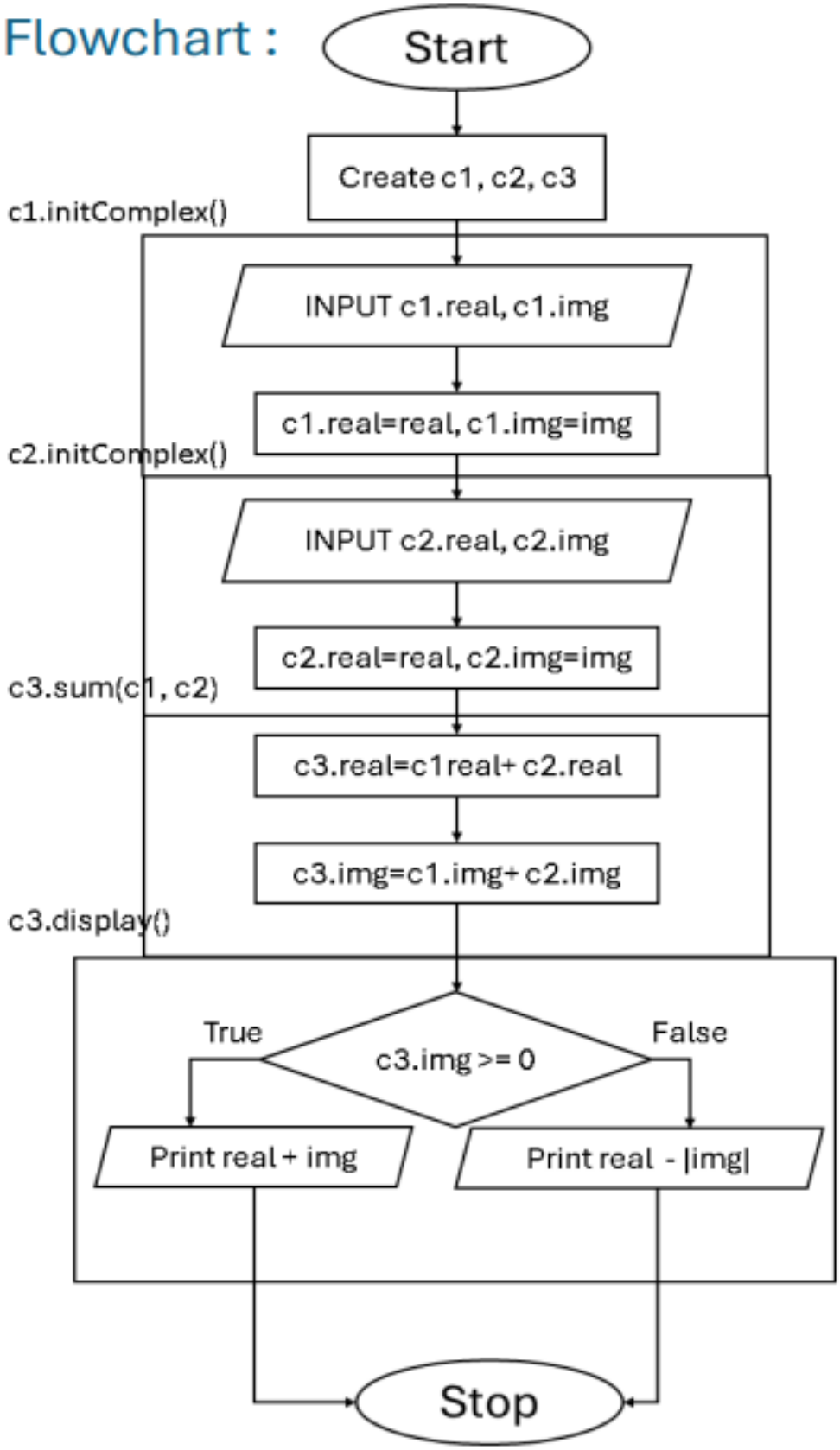
Step 2: Define a class or blueprint named Complex with two numeric variables: real and imag.

Step 3: Create three objects of the Complex class: c1 (first number), c2 (second number), and result (to store the sum).

Step 4: Read the input for the first complex number.

- Extract the first number and assign it to c1.real.
- Extract the second number and assign it to c1.imag.
- **Step 5:** Read the input for the second complex number.
- Extract the first number and assign it to c2.real.
- Extract the second number and assign it to c2.imag.
- **Step 6:** Compute the sum of the real parts.
- Calculate result.real = c1.real + c2.real.
- **Step 7:** Compute the sum of the imaginary parts.
- Calculate result.imag = c1.imag + c2.imag.
- **Step 8:** Format and display the output based on the value of result.imag.
- **If** result.imag is less than 0:
    - Print the format: result.real "-" Absolute(result.imag) "i"
    - Print the format: result.real "+" result.imag "i"
    - **Step 9:** Stop.

Flowchart :


Start
Create c1, c2, c3
c1.initComplex()
INPUT c1.real, c1.img
c1.real=real, c1.img=img
c2.initComplex()
INPUT c2.real, c2.img
c2.real=real, c2.img=img
c3.sum(c1, c2)
c3.real=c1real+ c2.real
c3.img=c1.img+ c2.img
c3.display()
True
c3.img >= 0
False
Print real + img
Print real - |img|
Stop

CODETANTRA Home

arjun.gahane.batch2025@sitnagpur.siu.edu.in Support Logout

14.1.1. Complex Number Addition using Class

04:08

Write a Python program using a class named **Complex** to add two complex numbers.

Create a class **Complex** with the following methods:

- **initComplex()**: Read user input for the real and imaginary parts.
- **sum()**: Compute the sum of two complex numbers and store the result in the current instance.
- **display()**: Display the complex number in the format "<a> + <b>i" or "<a> - <b>i".

The program should create three instances of **Complex** class: two for input complex numbers and one for the result. Initialize both complex numbers, compute their sum, and display the result.

Input Format:

- First line contains two space-separated numbers: real and imaginary parts of first complex number
- Second line contains two space-separated numbers: real and imaginary parts of second complex number

Output Format:

- Print a single line showing the sum of the two complex numbers in the format "<a> + <b>i" or " <a> - <b>i".

Sample Test Cases +

Explorer

complex...

Submit

```
class Complex:
# Constructor
    def __init__(self):
        self.real = 0
        self.imag = 0

    # Method to initialize complex number
    def initComplex(self):
        self.real, self.imag = map(int, input().split())

    # Method to compute sum
```

Debugger

Average time 0.011 s 10.63 ms

Maximum time 0.019 s 19.00 ms

4 out of 4 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Test case 1 19 ms Debug

Expected output	Actual output
3 4	3 4
2 5	2 5
5 + 9i	5 + 9i

Test case 2 9 ms

Terminal Test cases

< Prev Reset Submit Next >